Crystal Chiu

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Education

B.S. in Computer Science March 2025
University of California, Irvine

Experience

Independent Software Engineer | React Native, Expo, TypeScript, Supabase, Postgres Aug. 2026 - Present
CRIT Advisory — Contract *Remote*

- Provide full-stack services for an independent technology consulting business serving client needs across software and IT operations
- Client Engagement: Developing a high-customization digital library mobile app (DigiLib) for organizing, curating, and sharing collections of works of various formats including games, books, mangas, TV shows, and more
- Completed research, product design, and high-fidelity mockups, translating product vision into UIUX across core interactions while maintaining a distinct visual design
- Defined system architecture and technical specifications, including data modeling, backend architecture, API design for flexible and scalable media management, social features, and integrations with external APIs

Software Engineer | AWS, Docker, Laravel, Vue, React, Next, MySQL, Cursor Jul. 2025 - Jul. 2026
The Alliance Group Enterprise — Full Time *Santa Ana, CA*

- Migrated legacy infrastructure to AWS Cloud services with Terraform and Docker, transferring data seamlessly, increasing application capability, and improving deployment flexibility
- Worked with HR and accounting lead to build a custom timesheet service within the ERP system, cutting payroll overhead time, saving $100+/month, and providing real-time labor data
- Rebuilt a legacy project management frontend as a Vue SPA and refactored backend services into RESTful APIs, improving performance and mobile usability
- Reduced high-load API response times from 5–7s to 400–600ms through query optimization and Redis caching, easing database load
- Automated testing, image builds, and environment-specific deployments with GitHub Actions, replacing manual processes and increasing production uptime from 90% to 99.9%
- Standardized AI-assisted development with domain-specific context files, improving output consistency while reducing prompting, rework, and token usage by 25–35%

Student Software Developer | Laravel, Vue, Cypress, MySQL Mar. 2023 - Dec. 2024
UCI Student Affairs IT — Part-Time *Irvine, CA*

- Developed and maintained full-stack applications, primarily a commencement operations platform serving 10K+ students and staff, delivering features, bug fixes, and refactors through Agile development and quarterly release cycles
- Onboarded, trained, and mentored student developers through pair programming, code reviews, and guidance in navigating and debugging a large legacy codebase
- Worked on a dynamic seating chart tool for commencement organizers that automated seat assignments and reduced a weeks-long planning workflow

Projects

Popmart Restock Monitor | Node, MongoDB, Discord API

- Developed and deployed an API proxy scraper to track Popmart's "Labubu" product restocks and product changes, sending real-time Discord alerts to assist users in purchasing high-demand items with unpredictable availability
- Created multiple modes (Regular, Snooze, Throttle) for polite scraping to reduce unnecessary network traffic based on likelihood of restock time and priority items in the product line
- Reduced Puppeteer scrape times for JavaScript-heavy pages from 20s to 5s per page through targeted performance optimizations
- Implemented resilient error handling and health monitoring with automatic recovery and user alerts, using network congestion as a signal to detect potential product restocks

Skills

Languages: Java, JavaScript, TypeScript, Python, PHP, C++, SQL
Frameworks & Libraries: Vue.js, React.js, React Native, Node.js, Laravel, Flask
Databases & Data: MySQL, PostgreSQL, Redis
Cloud & DevOps: AWS, Docker, Terraform, GitHub Actions, CI/CD